

Assignment 2

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0.1 Instructions

This problem set is based on Lecture 7-11. The question are updated as we cover each lecture.

For each question:

1. Understand the economic intuition before using formulas.
2. Be clear about what the model is trying to answer.
3. Use concise reasoning.

1 Lecture 7: Interest-Rate Models

1.1 Question 1

A junior analyst says:

“Interest rates are uncertain, so we should use a short-rate model for every bond we price.”

The analyst is comparing the following securities:

Security	Description
A	5-year Treasury note with fixed annual coupons
B	5-year fixed-rate corporate bond with no embedded options
C	5-year callable agency bond
D	Interest-rate cap
E	Mortgage-backed security

Answer the following:

- Explain why Securities A and B usually do not require a stochastic short-rate model for time-0 valuation.
- State the basic valuation logic for a plain fixed-cash-flow bond using discount factors.
- Explain what a bootstrapped discount curve provides.
- Identify which securities in the table require future interest-rate dynamics and explain why.
- Correct the analyst's statement in one or two sentences.

1.2 Question 2

A portfolio manager asks why Lecture 7 spends time modeling the short rate if today's yield curve is already observable.

Answer the following:

- Explain the difference between a yield curve and a short-rate model.
- Explain what is meant by saying that a yield curve is a "snapshot."
- Explain what is meant by saying that a short-rate model is a "scenario generator."
- Explain why future discount curves $P(t, T)$ matter for securities whose cash flows depend on future rates.
- Give two examples of portfolio or risk-management questions that cannot be answered well using only today's yield curve.

1.3 Question 3

Consider the basic one-factor short-rate model:

$$dr_t = b(r_t, t)dt + \sigma(r_t, t)dz_t.$$

Answer the following:

- a. Explain the role of the drift term.
- b. Explain the role of the volatility term.
- c. Explain what mean reversion means and why it may be a reasonable assumption for interest rates.
- d. Compare the intuition behind constant volatility, level-proportional volatility, and square-root volatility.
- e. Explain one practical consequence of choosing a normal model that allows negative rates.

1.4 Question 4

A fixed-income team is deciding between an equilibrium model and an arbitrage-free model.

Answer the following:

- a. Explain the main purpose of an equilibrium interest-rate model.
- b. Explain the main purpose of an arbitrage-free interest-rate model.
- c. Explain why arbitrage-free models are usually preferred for pricing derivatives and bonds with embedded options.
- d. In the Hull-White model,

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t,$$

explain why $\theta(t)$ is central to fitting today's curve.

- e. Explain why a and σ are usually calibrated to interest-rate option prices rather than plain bond prices.

1.5 Question 5

An analyst wants to estimate historical short-rate volatility from weekly data. The analyst observes the following weekly changes in the short rate:

Week	Change in short rate
1	+4 bps
2	-2 bps
3	+6 bps
4	-5 bps
5	+3 bps
6	+1 bp
7	-4 bps
8	+5 bps

Answer the following:

- Explain why rate changes, rather than rate levels, are used to estimate volatility in this setting.
- Describe the steps for estimating weekly historical volatility from these observations.
- Compute the sample standard deviation of the weekly changes in basis points.
- Annualize the weekly volatility using $\sqrt{52}$.
- Explain two limitations of using this historical volatility estimate as an input to a short-rate model.